

Let the World's Leading Expert Show You Their Table of Contents

Using Claude to satisfy conditional parallel trends
in difference-in-differences

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Baylor University · Sant'Anna Workshop on Causal Inference

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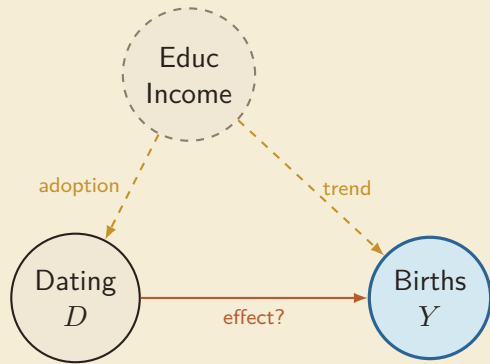
Online dating spread unevenly across U.S. counties, 1995–2007

- Broadband rollout + Match.com (1995), eHarmony (2000), OkCupid (2004)
- Urban, educated, younger counties adopted first
- Research question: did online dating *change* birth rates?
- Natural strategy:
difference-in-differences
 - Early-adopter vs. late-adopter counties
 - Pre- vs. post-broadband rollout

$$\hat{\tau} = \underbrace{(\bar{Y}_{T,\text{post}} - \bar{Y}_{T,\text{pre}})}_{\text{treated change}} - \underbrace{(\bar{Y}_{C,\text{post}} - \bar{Y}_{C,\text{pre}})}_{\text{control change}}$$

Valid only if trends would have been equal without treatment

The challenge: adoption correlates with everything that predicts birth rates



If X causes both adoption *and* the pre-treatment trend in birth rates, treated and control counties are on **different trajectories** before treatment. The DiD estimate is biased.

Urban, high-education, high-income counties adopted broadband first — the same counties where birth rates were already falling due to female LFP growth. Without conditioning on X , we are comparing apples to oranges.

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Act II: The Fix

Conditional Parallel Trends

Heckman, Ichimura & Todd (1997): condition on the right X

Unconditional Parallel Trends

$$\mathbb{E}[\Delta Y(0) \mid D = 1] = \mathbb{E}[\Delta Y(0) \mid D = 0]$$

Treated and control would have changed by the same amount.

Conditional Parallel Trends (Heckman, Ichimura & Todd 1997)

$$\mathbb{E}[\Delta Y(0) \mid D = 1, X = x] = \mathbb{E}[\Delta Y(0) \mid D = 0, X = x] \quad \forall x \in \mathcal{X}$$

Within cells defined by X , the counterfactual trend is the same.

Parallel trends holds *after* conditioning on covariates that drive both selection and outcome trends. X does the work.

The potential-outcomes framing makes the identification crystal clear

The setup

- $Y_{it}(1)$: birth rate if county i adopts
- $Y_{it}(0)$: birth rate if not
- $D_i = 1$ if early-adopter county
- $\tau_i = Y_{it}(1) - Y_{it}(0)$ (unobserved)

Target estimand

$$\text{ATT} = \mathbb{E}[\tau_i \mid D_i = 1]$$

The key insight

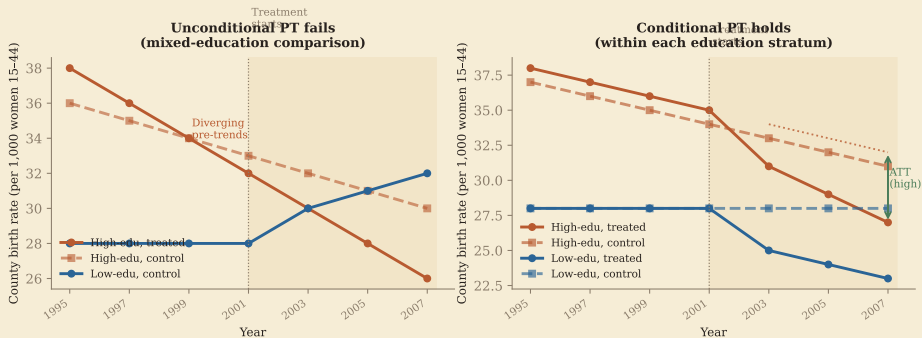
ATT requires the counterfactual trend:
what would treated counties have done without treatment?

Controls tell us — but only if their trend matches treated **within strata of X** .

The whole game is finding the right X .

Within strata, parallel trends is restored

Conditioning on covariates rescues parallel trends



Left: Pooling mixed-education counties creates diverging pre-trends. Right: Within each education stratum, treated and control trend together. Conditioning on X makes DiD valid.

So what belongs in X ? — This is the hard part

X should contain every county-level characteristic that:

- (a) predicts *adoption of online dating*, and
- (b) drives the *trend* in birth rates over 1995–2007.

Get (a) wrong: confounding. Get (b) wrong: differential pre-trends.

Wrong approach:

- Add controls “because they matter”
- Controls chosen to stabilize coefficients
- Post-treatment controls
- Kitchen-sink regressions

Right approach:

- Map the DGP of $Y(0)$
- What drives birth rate *trends*?
- Same variables predict adoption
- Document the reasoning

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Act III: The Insight

Ask the World's Leading Expert

The expert shortcut: who already mapped the DGP of $Y(0)$?

The DGP of county-level birth rates in America has been studied for decades.

Someone has written the **definitive book** on it.

That book's table of contents *is* a covariate list.

Every chapter title answers: "What drives $Y(0)$?"

We just need to ask the right expert.

Expert's Book

Ch. 1: Female Education

Ch. 2: Labor

Force Participation

Ch. 3: Welfare &
Transfer Payments

⋮

Ch. 20: Pre-Treatment Trends

← This is X .

The Claude prompt: role-play as the world's leading expert

The prompt

You are Professor Elena Marchetti, University of Michigan, author of *The American Fertility Landscape: County-Level Determinants of Birth Rates in the Postindustrial Era* (Princeton UP, 2020).

I am studying the effect of online dating on U.S. birth rates using county-level DiD, 1995--2007, and need to satisfy conditional parallel trends. Please give me your book's complete table of contents. Each chapter title should name a determinant of county birth rates. I will use this as my covariate list.

The role-play forces Claude to *commit* to a theory of the DGP: not “things that might matter” but a structured, chapter-by-chapter account of what drives $Y(0)$.

Professor Elena Marchetti's definitive account of American fertility

The American Fertility Landscape County-Level Determinants of Birth Rates in the Postindustrial Era

Elena Marchetti · University of Michigan

Princeton University Press, 2020 · *PAA Book Award*

Prof. Marchetti's field — formal demography and applied fertility economics — maps exactly onto the DGP of $Y(0)$. Her book's chapter structure is the theory of county birth rate trends.

The table of contents, Part I: Economic and Social Foundations

Ch. 1 Female Education and Fertility Timing

Ch. 2 Female Labor Force Participation as Opportunity Cost

Ch. 3 Welfare, TANF, and Fertility Incentives

Ch. 4 Hispanic Migration, Cultural Norms, and Fertility

Ch. 5 Marriage Markets and Partnership Formation

Each chapter title encodes a theory of what drives $Y(0)$: education changes the timing of fertility; female LFP raises its opportunity cost; welfare creates margin-altering incentives; immigration shifts cultural norms around family size; marriage markets set the partnership context for birth.

The table of contents, Part II: Biology, Geography, and Behavior

Ch. 6 Teen Birth Rates as Baseline Fertility Measure

Ch. 7 Religious Adherence and Fertility Norms

Ch. 8 Rural–Urban Geography and Family Structure

Ch. 9 Population Age Structure and Reproductive Exposure

Ch. 10 Housing Costs and the Economics of Family Formation

Prior teen birth rates serve as the pre-treatment baseline and placebo. Rural geography predicts both late broadband adoption and higher baseline fertility. Age structure determines the share of women in peak reproductive years. Housing costs constrain family formation decisions.

The table of contents, Part III: Labor Markets and Healthcare

Ch. 11 Industrial Structure, Deindustrialization, and Fertility

Ch. 12 Unemployment, Labor Demand, and Birth Timing

Ch. 13 Health Insurance Coverage and Contraceptive Access

Ch. 14 Abortion Access, Provider Availability, and Fertility

Ch. 15 Poverty, Median Income, and Fertility Timing

Deindustrialization, unemployment, and income jointly shape the economic context for fertility decisions. Health insurance coverage determines contraceptive access. Abortion provider availability is a direct constraint on realized birth rates.

The table of contents, Part IV: Demographics, Policy, and History

Ch. 16 Military Employment and Family Formation Incentives

Ch. 17 Substance Use, Health Risk, and Unintended Birth Rates

Ch. 18 Racial and Ethnic Composition and Fertility Norms

Ch. 19 State Policy Environments and Family Formation

Ch. 20 Pre-Treatment Fertility Trends and Parallel Trends Validation

Military counties have distinctive fertility incentives. Substance use correlates with unintended births. Racial composition shapes cultural fertility norms. Chapter 20 is explicitly the parallel-trends validation chapter: the pre-treatment trend *is* a covariate.

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Act IV: From Theory to Data

Walking All Twenty Chapters

Selection criteria: theory precedes download

Three requirements for every dataset:

1. **County-level.** State-level averages mask the variation DiD needs.
2. **Annual, 1995–2007.** Must span the treatment window; gaps introduce differential attrition bias.
3. **Publicly available.** Restricted-use data is fine to note, but the replication archive must be constructable from public sources.

Sources by reliability tier:

- A BLS LAUS, BEA Regional, SEER, NCHS natality — annual, consistent methodology, machine-readable
- B Census SAIPE, FHFA HPI, Census intercensal — excellent but occasional gaps
- C Guttmacher, USDA, NCES — specialized, verify carefully

Chapters 1–5: Social and economic foundations

Ch.	Dimension	Data sources
1	Female education & fertility timing	Census intercensal estimates (age/sex); NCES CCD county data
2	Female labor force participation	BLS LAUS county LFP 1995–2007; BEA CAINC4 wages by industry
3	Welfare, TANF, fertility incentives	BEA CAINC35 government transfer payments (county)
4	Hispanic migration & cultural norms	Census intercensal race/Hispanic estimates; BEA CAINC5N
5	Marriage markets	BEA CAINC1 county personal income; NCHS vital statistics

Gold standard for Ch. 1/5: Census 2000 SF3 educational attainment (B15002) and marital status (B12001) — free key at api.census.gov.

Chapters 6–10: Biology, geography, and housing

Ch.	Dimension	Data sources
6	Teen birth rates (baseline)	NCHS natality microdata 1995, 1997, 2003, 2007 (county FIPS)
7	Religious adherence & norms	USDA Rural Atlas county data; USDA RUCC 2003 rural codes
8	Rural–urban classification	USDA Rural-Urban Continuum Codes 1993 and 2003
9	Population age structure	NCI SEER county pop. by age/sex 1990–2024 (91 MB); BLS QCEW
10	Housing costs	FHFA HPI county annual; BEA CAINC1 personal income

Ch. 7: Gold standard is ARDA/Glenmary RCMS 2000 county church membership (proxy used here). *SEER:*

Fixed-width; parse with `gzip -dc | awk` per NCI layout.

Chapters 11–15: Labor markets, healthcare, poverty

Ch.	Dimension	Data sources
11	Industry structure & deindustrializ.	BEA CAINC4 wages; BEA CAINC5N personal income; BLS QCEW
12	Unemployment & labor demand	BLS LAUS county annual 1995–2007 (13 files, consistent series)
13	Health insurance coverage	Census SAHIE 2008–2012 county (ACS-based; proxy for pre-period)
14	Abortion access	Guttmacher provider locator; BEA CAINC35 transfer payments
15	Poverty & income	Census SAIPE county poverty/income 1995–2007; BEA CAINC1

SAHIE: ACS-based county estimates begin 2008; use as post-period baseline. *SAIPE*: Most reliable county poverty series; covers all years 1995–2007.

Chapters 16–20: Military, risk, demographics, policy, trends

Ch.	Dimension	Data sources
16	Military employment	BEA CAINC30 federal/military pay; BLS QCEW county industry
17	Substance use & health risk	BEA CAINC91 health expenditures; Census SAIPE poverty proxy
18	Racial/ethnic composition	NCI SEER race-stratified pop.; Census intercensal race/Hispanic
19	State policy environment	BEA SAGDP2N state GDP; BEA SAINC25 social insurance receipts
20	Pre-treatment fertility trends	NCHS natality 1995 and 2000 microdata (county birth rates)

Ch. 17: Gold standard: SAMHSA TEDS county treatment admissions (registration required). *Ch. 20 is structural:* pre-treatment birth rate trend is the most important covariate for parallel trends validation — and the most difficult to fabricate.

What we got vs. what we ideally want

We have (public):

- BLS LAUS: unemployment/LFP 1995–2007
- BEA CAINC 1/4/5N/30/35/91: income, wages
- NCI SEER: pop. by age/sex/race 1990–2024
- NCHS natality: birth microdata, county FIPS
- Census SAIGE: poverty/income 1995–2007
- FHFA HPI: county house prices
- Census intercensal: race/Hispanic 2000–10

68 files, 1.3 GB across 20 chapter directories. 18 of 20 chapters covered with good public proxies. All gaps documented.

Still needed:

- **Census API key:** college (B15002), Hispanic (B03003), marital (B12001)
- **ARDA RCMS:** county church membership
- **SAMHSA TEDS:** substance use admissions
- **SAHIE pre-2008:** health insurance
- Military base GIS for county assignment

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Act V: Honest Reckoning

What a Skeptic Would Say

A skeptic would say: isn't this just a very fancy kitchen sink?

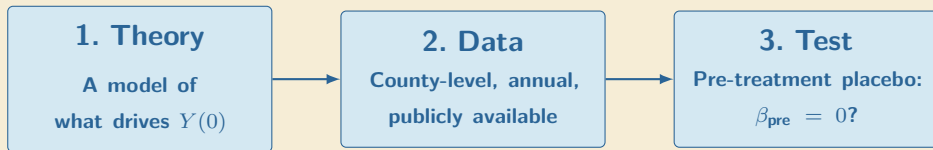
The critique

“You added 20 covariate dimensions because a language model playing a fictional professor told you to. You have no principled argument for why these satisfy conditional parallel trends.”

The response

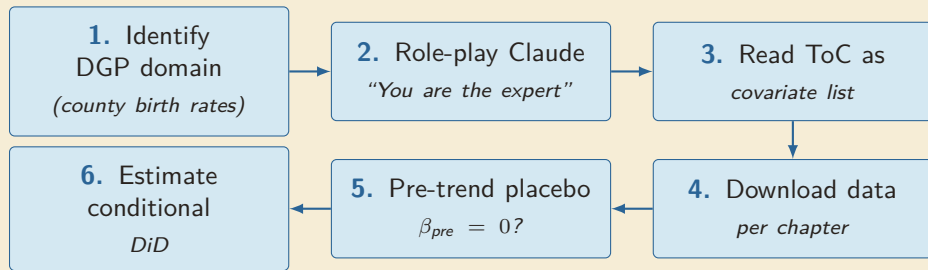
- The ToC encodes a *theory of the DGP* grounded in demography.
- We can test it: run pre-treatment regressions. If X satisfies conditional PT, $\hat{\beta}_{\text{pre}} = 0$.
- The exercise forces us to *commit* before observing post-treatment data. That discipline is the point.
- Selection that amounts to “stabilize my coefficient” is *worse*, not better, than a theory-grounded list.

Three things conditional parallel trends actually requires



Claude does step 1 faster. Step 2 is where you spend a weekend. Step 3 is where you find out if you were right.

The workflow: from fiction to falsifiable covariate list



*If step 5 fails: return to step 3.
If it passes: trust your DiD.*

**Let the world's leading expert on $Y^{(0)}$
show you their table of contents.**

That's your covariate list.

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